



KUWAIT UNIVERSITY
Faculty of Science
Department of Mathematics

Math 101: Calculus I

First Exam, 14 November 2017

Duration: 90 minutes

- *In questions 1-8, show all your work.*
- *Calculators and communication devices are not allowed in the examination room.*

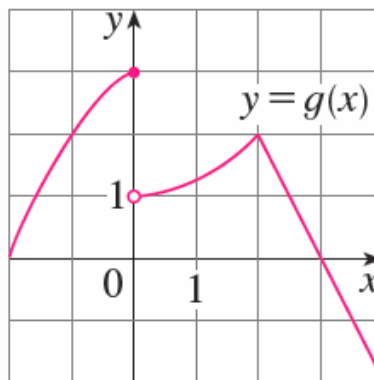
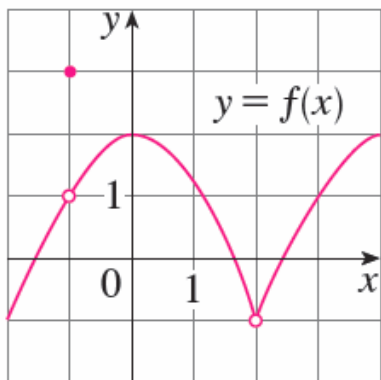
Name: _____

Student ID: _____

Section Number: _____ Serial Number: _____

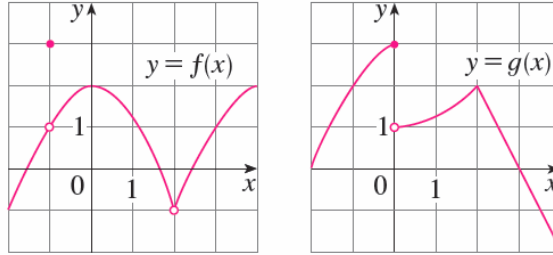
Q1 (20 pts)	
Q2 (20 pts)	
Q3 (10 pts)	
Q4 (10 pts)	
Q5 (10 pts)	
Q6 (10 pts)	
Q7 (10 pts)	
Q8 (10 pts)	
Total	

1. [5 + 5 + 10 = 20 pts.] Use the given graphs of f and g to find each of the following limits, if it exists.



- (a) $\lim_{x \rightarrow 2} [f(x) + g(x)] =$
- (b) $\lim_{x \rightarrow 3} \frac{g(x)}{f(x)} =$
- (c) $\lim_{x \rightarrow 0} [f(x)g(x)] =$
2. [10+10=20 pts.] Let $f(x) = \frac{3x^2 + 10}{x\sqrt{x^2 + 1}}$.
- (a) Find the horizontal asymptotes of the graph of f , if any.
- (b) Find the vertical asymptotes of the graph of f , if any.
3. [10 pts.] Let $f(x) = \ln\left(\frac{x}{2x-1}\right)$. Find $f^{-1}(x)$.
4. [5 + 5 = 10 pts.] Let $f(x) = \frac{\sin(x-1)}{x^3-1}$.
Show that f is discontinuous at $x = 1$, and classify the discontinuity as removable, jump, or infinite.
5. [10 pts.] Let $f(x) = \frac{2}{x+1}$. Use the definition of the derivative to find $f'(0)$.
6. [10 pts.] Use the Intermediate Value Theorem to show that the equation $\cos x - \ln(x+1) = 0$ has a real root.
7. [5 + 5 = 10 pts.] Let $f(x) = e^x - 2x$.
- (a) Find the point on the curve of f where the tangent line is horizontal.
- (b) Using part (a) find an equation of this horizontal tangent line.
8. [10 pts.] Let $f(x) = x^2 \tan^3(2x)$. Find $f'(x)$.

1. [5 + 5 + 10 = 20 pts.] Use the given graphs of f and g to find each of the following limits, if it exists.



(a) $\lim_{x \rightarrow 2} [f(x) + g(x)] = \lim_{x \rightarrow 2} f(x) + \lim_{x \rightarrow 2} g(x) = -1 + 2 = 1.$

(b) $\lim_{x \rightarrow 3} \frac{g(x)}{f(x)} = \frac{\lim_{x \rightarrow 3} g(x)}{\lim_{x \rightarrow 3} f(x)} = \frac{0}{1} = 0.$

(c) $\lim_{x \rightarrow 0} [f(x)g(x)] = \text{DNE.} \quad (2 \text{ pts})$

Since $\lim_{x \rightarrow 0^+} [f(x)g(x)] = \lim_{x \rightarrow 0^+} f(x) \lim_{x \rightarrow 0^+} g(x) = (2)(1) = 2 \quad (4 \text{ pts})$
 and $\lim_{x \rightarrow 0^-} [f(x)g(x)] = \lim_{x \rightarrow 0^-} f(x) \lim_{x \rightarrow 0^-} g(x) = (2)(3) = 6. \quad (4 \text{ pts})$

2. [10+10=20 pts.] Let $f(x) = \frac{3x^2 + 10}{x\sqrt{x^2 + 1}}.$

- (a) Find the horizontal asymptotes of the graph of f , if any.

Since $\lim_{x \rightarrow \pm\infty} \frac{3x^2 + 10}{x\sqrt{x^2 + 1}} = \lim_{x \rightarrow \pm\infty} \frac{x^2(3 + 10/x^2)}{x|x|\sqrt{1 + 1/x^2}} = \lim_{x \rightarrow \pm\infty} \pm \frac{x^2(3 + 10/x^2)}{x^2\sqrt{1 + 1/x^2}} = \boxed{\pm 3}.$

Therefore, the lines $y = 3$ and $y = -3$ are horizontal asymptotes of the graph of f .

- (b) Find the vertical asymptotes of the graph of f , if any.

Since $\lim_{x \rightarrow 0^\pm} \frac{3x^2 + 10}{x\sqrt{x^2 + 1}} = \boxed{\pm\infty}.$

Therefore, the line $x = 0$ is a vertical asymptote of the graph of f .

3. [10 pts.] Let $f(x) = \ln\left(\frac{x}{2x-1}\right).$ Find $f^{-1}(x).$

(1) Let $y = \ln\left(\frac{x}{2x-1}\right).$

(2) Solve for x in terms of y : $x = \frac{e^y}{2e^y - 1}.$

(3) Interchange x and y : $y = \frac{e^x}{2e^x - 1}.$ Therefore, $f^{-1}(x) = \frac{e^x}{2e^x - 1}.$

4. [5 + 5 = 10 pts.] Let $f(x) = \frac{\sin(x-1)}{x^3-1}$.

Show that f is discontinuous at $x = 1$, and classify the discontinuity as removable, jump, or infinite.

We have $\lim_{x \rightarrow 1} \frac{\sin(x-1)}{x^3-1} = \lim_{x \rightarrow 1} \frac{\sin(x-1)}{(x-1)(x^2+x+1)} = \lim_{x \rightarrow 1} \frac{\sin(x-1)}{(x-1)} \lim_{x \rightarrow 1} \frac{1}{(x^2+x+1)} = \boxed{1/3}$.

Since f is undefined at $x = 1$, therefore f has a removable discontinuity at 1.

5. [10 pts.] Let $f(x) = \frac{2}{x+1}$. Use the definition of the derivative to find $f'(0)$.

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{\frac{2}{x+1} - 2}{x} = \lim_{x \rightarrow 0} \frac{2 - 2x - 2}{x(x+1)} = \lim_{x \rightarrow 0} \frac{-2x}{x(x+1)} = -2.$$

6. [10 pts.] Use the Intermediate Value Theorem to show that the equation $\cos x - \ln(x+1) = 0$ has a real root.

$$\boxed{\text{Let } f(x) = \cos x - \ln(x+1)}.$$

The function f is continuous on its domain $(-1, \infty)$, hence its continuous on $[0, \pi]$. Since $f(0) = 1 > 0$ and $f(\pi) = -1 - \ln(\pi+1) < 0$, therefore by the IVT there is a number $c \in (0, \pi)$ s.t. $f(c) = 0$. Hence, the given equation has a real root.

7. [5 + 5 = 10 pts.] Let $f(x) = e^x - 2x$.

(a) Find the point on the curve of f where the tangent line is horizontal.

We have $\boxed{f'(x) = e^x - 2}$.

Thus, $\boxed{f'(x) = 0}$ when $e^x = 2$ i.e., $\boxed{x = \ln 2}$. So the given curve has horizontal tangent when $\boxed{x = \ln 2}$. The corresponding point is $\boxed{(\ln 2, 2 - 2 \ln 2)}$.

(b) Using part (a) find an equation of this horizontal tangent line.

Equation of this tangent line is $\boxed{y = 2 - 2 \ln 2}$.

8. [10 pts.] Let $f(x) = x^2 \tan^3(2x)$. Find $f'(x)$.

$$\boxed{f'(x) = 2x \tan^3(2x) + 6x^2 \tan^2(2x) \sec^2(2x)}.$$